

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 3, Randomized, Double-blind Trial of Pembrolizumab (MK-3475) Plus Enzalutamide Plus ADT Versus Placebo Plus Enzalutamide Plus ADT in Participants With Metastatic Hormone-Sensitive Prostate Cancer (mHSPC) (KEYNOTE-991)
Short Title:	Efficacy and Safety of Pembrolizumab (MK-3475) Plus Enzalutamide Plus Androgen Deprivation Therapy (ADT) Versus Placebo Plus Enzalutamide Plus ADT in Participants With Metastatic Hormone-Sensitive Prostate Cancer (mHSPC) (MK-3475-991/KEYNOTE-991)-China Extension
Protocol Number:	853812311392007606
NCT Number:	NCT04934722
Sponsor Name:	Merck
Investigational Product:	enzalutamide
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Radiographic Progression-free Survival (rPFS) Per Prostate Cancer Working Group (PCWG)-Modified Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) as Assessed by Blinded Independent Central Review (BICR): rPFS was defined as the time from randomization to occurrence of: radiological tumor progression using RECIST 1.1 as assessed by BICR; progression of bone lesions using PCWG criteria; or death due to any cause. Progression as per modified RECIST 1.1 was ≥20% increase in sum of diameters of target lesions and progression of existing non-target lesions. Progression of bone lesions by PCWG criteria was the appearance of ≥2 new bone lesions on bone scan, that have been confirmed to not represent tumor flare, and was persistent for ≥6 weeks. The rPFS was calculated using the product-limit (Kaplan-Meier) method for censored data. Participants without a rPFS event were censored at the date of last disease assessment. 2. Overall Survival (OS): OS was defined as the time from randomization to death due to any cause. The OS was calculated using the product-limit (Kaplan-Meier) method for censored data. Participants without documented death at the time of the analysis were censored at the date of the last follow-up.
Secondary Obj.:	1. Time to Initiation of the First Subsequent Anti-cancer Therapy or Death (TFST) 2. Time to First Symptomatic Skeletal-related Event (TTSSRE) 3. Time to Prostate-specific Antigen (PSA) Progression

2.2 Background & Rationale

This study will assess the efficacy and safety of pembrolizumab plus enzalutamide plus ADT versus placebo plus

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enzalutamide plus ADT in Chinese participants with mHSPC. The primary hypothesis is that in participants with mHSPC, the combination of pembrolizumab plus enzalutamide plus ADT is superior to placebo plus enzalutamide plus ADT with respect to 1) radiographic progression-free survival (rPFS) per Prostate Cancer Working Group (PCWG)-modified Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 as assessed by blinded independent central review (BICR) and 2) overall survival (OS). As of Amendment 4, the study is being stopped for futility. All the prespecified interim analysis after interim analysis (IA1) and final analysis of the study described the statistical analysis plan (SAP) will not be performed. Safety analysis will be performed at the end of the study; there will be no further analyses for efficacy and electronic patient-reported outcome (ePRO) endpoints collected from participants beyond the IA1 cutoff date. All study participants will stop ongoing treatment with pembrolizumab/placebo. Exceptions may be requested for study participants who, in the assessment of their study physician, are benefitting from the combination of enzalutamide and pembrolizumab, after consulting with the Sponsor. All other study participants should be discontinued from study and be offered standard of care (SOC) treatment as deemed necessary by the Investigator. If enzalutamide as SOC is not accessible off study to the participant, central sourcing may continue. As of Amendment 04, disease progression will no longer be centrally verified, participants will only be assessed locally. As of Amendment 4, Second Course treatment is not an option for participants. There are currently no participants in the Second Course Phase.

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE3, Status: COMPLETED

Target N: 186

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- * Male participants with histologically- or cytologically-confirmed adenocarcinoma of the prostate without small cell histology
- * Has metastatic disease assessed by investigator and verified by BICR by either ≥2 bone lesions on bone scan and/or visceral disease by computed tomography/magnetic resonance imaging (CT/MRI)
- * Willing to maintain continuous Androgen Deprivation Therapy (ADT) with a luteinizing-hormone releasing hormone (LHRH) agonists or antagonists during study treatment or have a history of bilateral orchiectomy
- * Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 assessed within 10 days of randomization
- * Participants receiving bone resorptive therapy (including, but not limited to, bisphosphonate or denosumab) must have been on stable doses prior to randomization
- * Has adequate organ function
- * Has provided newly obtained core or excisional biopsy (obtained within 12 months of screening) from soft tissue not previously irradiated (samples from tumors progressing in a prior site of radiation are allowed). Participants with bone only or bone predominant disease may provide a bone biopsy sample
- * Male participants must agree to the following during the intervention period and for at least 120 days after the last dose of study intervention: Refrain from donating sperm PLUS either be abstinent from heterosexual intercourse and agree to remain abstinent OR agree to use contraception, unless confirmed to be azoospermic
- * Male participants must agree to use male condom when engaging in any activity that allows for passage of ejaculate to

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another person of any sex

4.2 Exclusion Criteria

- * Has a known additional malignancy that is progressing or has required active treatment in the last 3 years
- * Has an active autoimmune disease that has required systemic treatment in past 2 years
- * Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy
- * Has undergone major surgery including local prostate intervention (excluding prostate biopsy) within 28 days prior to randomization and not recovered adequately from the toxicities and/or complications
- * Has a gastrointestinal disorder affecting absorption or is unable to swallow tablets/capsules
- * Has an active infection (including tuberculosis) requiring systemic therapy
- * Has a history of (non-infectious) pneumonitis that required steroids or has current pneumonitis
- * Has known active human immunodeficiency virus (HIV), hepatitis B virus (HBV) or hepatitis C virus (HCV) infection
- * Has known or suspected central nervous system (CNS) metastases and/or carcinomatous meningitis
- * Has a history of seizure or any condition that may predispose to seizure
- * Has a history of loss of consciousness within 12 months of screening
- * Has had myocardial infarction or uncontrolled angina within 6 months prior to randomization, or has New York Heart Association class III or IV congestive heart failure or a history of New York Heart Association class III or IV congestive heart failure
- * Has hypotension (systolic blood pressure $\leq$ 86 millimeters of mercury [mmHg]) or uncontrolled hypertension (systolic blood pressure $\geq$ 170 mmHg or diastolic blood pressure $\geq$ 105 mmHg) at the screening visit
- * Has a history of clinically significant ventricular arrhythmias
- * Has hypersensitivity to pembrolizumab and/or enzalutamide and/or any of their excipients
- * Has received prior ADT as neoadjuvant/adjuvant therapy for non-metastatic prostate cancer for $\geq$ 39 months in duration or within 9 months prior to randomization or with evidence of disease progression while receiving ADT
- * Has had prior treatment with a next generation hormonal agent (eg, abiraterone, enzalutamide, apalutamide, darolutamide)
- * Has received prior therapy with an anti-programmed cell death-1 (anti-PD-1), anti-programmed cell death-ligand 1 (anti-PD-L1), or anti-programmed cell death-ligand 2 (anti PD-L2) agent or with an agent directed to another stimulatory or coinhibitory T-cell receptor
- * Has received a live vaccine within 30 days prior to randomization
- * Has a "superscan" bone scan
- * Has had an allogenic tissue/solid organ transplant
- * Is expecting to conceive or father children within the projected duration of the study, starting with the screening visit through 120 days after the last dose of study treatment
- * Has received any prior pharmacotherapy, radiation therapy or surgery for metastatic prostate cancer with the following exceptions:
 1. Up to 3 months of ADT or orchiectomy with or without concurrent first-generation antiandrogens, if patient was not treated with docetaxel
 2. May have 1 course of palliative radiation or surgical therapy to treat symptoms resulting from metastatic disease if it was administered at least 4 weeks prior to randomization

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3. For participants with low volume metastatic disease, may have 1 course of definitive radiotherapy if it was administered at least 4 weeks prior to randomization

4. Up to 6 cycles of docetaxel therapy with final treatment administration completed within 2 months of randomization and no evidence of disease progression. In these participants up to 6 months of ADT permitted

11. Study Identifiers & Governance

Primary ID: 853812311392007606

Registry Number: NCT04934722

Sponsor: Merck

Study Title: Efficacy and Safety of Pembrolizumab (MK-3475) Plus Enzalutamide Plus Androgen Deprivation Therapy (ADT) Versus Placebo Plus Enzalutamide Plus ADT in Participants With Metastatic Hormone-Sensitive Prostate Cancer (mHSPC) (MK-3475-991/KEYNOTE-991)-China Extension

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____